

Cloud Security Assessment Report

Google Cloud IAM Security Implementation and Access Control Validation

Report Information

Project Title

GCP IAM Security Lab: Implementing Least-Privilege Access Control with Service Accounts and Custom IAM Roles

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Role

Future Cloud Security Engineer

Report Type

Cloud Security Assessment Report

Environment

Google Cloud Platform (GCP)

Security Domain

Identity and Access Management (IAM)

Assessment Date

June 2026

Executive Summary

A cloud security assessment was conducted within a Google Cloud Platform (GCP) environment to evaluate Identity and Access Management (IAM) controls and validate secure access management practices.

The assessment focused on implementing least-privilege access controls through service accounts, predefined IAM roles, custom IAM roles, and policy verification using both the Google Cloud Console and gcloud command-line tools.

The objective was to establish a secure access control model that minimizes excessive permissions, reduces the risk of privilege escalation, and aligns with cloud security governance best practices.

The assessment successfully demonstrated secure identity management, role-based access control (RBAC), custom role development, policy validation, and resource cleanup procedures.

Section 1: Identify the Cloud Security Control Implemented

The primary security control implemented during this project was Google Cloud Identity and Access Management (IAM).

IAM provides centralized control over authentication and authorization within Google Cloud Platform by defining which identities can access specific resources and which actions they are permitted to perform.

The assessment focused on the following IAM security mechanisms:

- Service Accounts
- Role-Based Access Control (RBAC)
- Predefined IAM Roles
- Custom IAM Roles
- IAM Policy Bindings
- Least Privilege Access Management

A dedicated service account was created and assigned only the permissions required to perform specific Compute Engine operations.

Instead of granting broad administrative privileges, a custom IAM role was developed containing only the following permissions:

- compute.instances.list
- compute.instances.start
- compute.instances.stop

This implementation demonstrates the Principle of Least Privilege by restricting access to only necessary operations.

Section 2: Cloud Security Implementation and Analysis

A dedicated Google Cloud project named:

`gcp-iam-security-lab`

was created to establish an isolated environment for IAM configuration and testing.

The cloud security analyst enabled billing services and activated the Compute Engine API to support IAM permission management and access validation activities.

A service account named:

`cloud-security-viewer`

was created to represent a non-human identity commonly used by applications and automation workflows.

The service account was assigned the Viewer role to provide read-only access to cloud resources while preventing unauthorized modifications.

To demonstrate granular permission management, the analyst developed a custom IAM role using a JSON role definition file and deployed the role using the `gcloud` command-line interface.

The custom role contained only three permissions:

- `compute.instances.list`
- `compute.instances.start`
- `compute.instances.stop`

The role was assigned to the service account through IAM policy bindings.

Following deployment, policy verification activities were performed using `gcloud` commands to inspect active IAM bindings and confirm successful role assignment.

Verification results confirmed that the service account received both the Viewer role and the custom IAM role.

Throughout the implementation, no service account keys were generated, reducing credential exposure risks and aligning with cloud security best practices.

The assessment concluded that access controls were functioning as intended and that least-privilege principles were successfully enforced.

Section 3: Security Risks Identified

The assessment identified several cloud security risks commonly associated with misconfigured IAM environments.

Risk 1: Excessive Permissions

Granting broad administrative roles may allow unauthorized resource modifications and increase the likelihood of privilege escalation.

Risk 2: Service Account Abuse

Improperly managed service accounts may provide attackers with persistent access to cloud resources.

Risk 3: Credential Exposure

Service account keys stored insecurely can be abused to gain unauthorized access.

Risk 4: Permission Creep

Users and service accounts often accumulate unnecessary permissions over time.

Risk 5: Misconfigured Access Controls

Incorrect IAM policy assignments may expose sensitive resources to unauthorized users.

Section 4: Security Controls Implemented

The following controls were implemented to mitigate identified risks:

Principle of Least Privilege (PoLP)

Permissions were restricted to only those required for operational functionality.

Role-Based Access Control (RBAC)

Access was controlled through predefined and custom IAM roles.

Custom IAM Roles

Granular permissions replaced broad administrative privileges.

Service Account Security

Dedicated service accounts were used instead of assigning permissions directly to users.

Keyless Authentication

No service account keys were created.

IAM Policy Verification

Policy bindings were validated using command-line inspection.

Secure Resource Cleanup

All IAM resources were removed after project completion.

Section 5: Evidence Collected

The following evidence was collected during the assessment:

Evidence ID	Description
E-01	GCP Project Creation
E-02	Billing Configuration
E-03	Compute Engine API Enablement
E-04	Service Account Creation
E-05	IAM Role Assignment
E-06	Custom IAM Role Creation
E-07	Custom Role Assignment
E-08	IAM Policy Verification
E-09	Project Completion

Section 6: Security Recommendations

The following recommendations are advised for production cloud environments:

Enforce Multi-Factor Authentication (MFA)

Require MFA for all privileged cloud accounts.

Implement Periodic IAM Reviews

Conduct regular access reviews to remove unnecessary permissions.

Enable Cloud Audit Logging

Monitor IAM activities and administrative actions.

Restrict Service Account Key Usage

Adopt keyless authentication whenever possible.

Implement Organization Policies

Enforce centralized governance controls across cloud projects.

Apply Separation of Duties

Prevent a single identity from having unrestricted administrative access.

Use Custom Roles

Replace broad predefined roles with task-specific permissions.

Business Impact

Strong IAM controls significantly reduce the likelihood of cloud security incidents caused by excessive permissions, unauthorized access, and identity misuse.

The implementation provides:

- Improved access governance
 - Reduced attack surface
 - Enhanced accountability
 - Better compliance readiness
 - Stronger cloud security posture
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Conclusion

This cloud security assessment successfully demonstrated the implementation of secure Identity and Access Management controls within Google Cloud Platform.

The project validated the use of service accounts, role-based access control, custom IAM roles, and policy verification procedures while enforcing the Principle of Least Privilege.

The assessment confirms that properly configured IAM controls are essential for protecting cloud resources, preventing unauthorized access, and maintaining a secure cloud operating environment.

The skills demonstrated throughout this assessment closely align with real-world responsibilities performed by Cloud Security Engineers responsible for securing enterprise cloud infrastructures.